

Experiment - 7.1.2. Possible Combinations from Three Digits

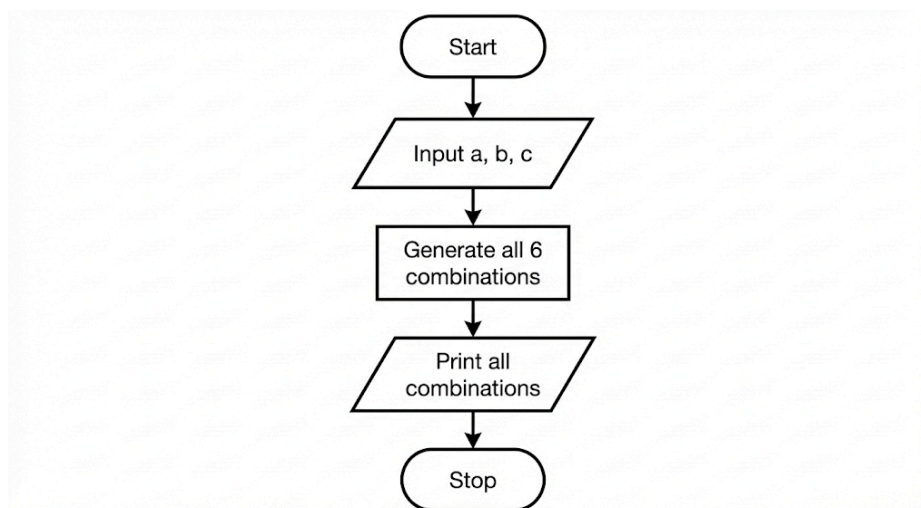
1. Aim

To design and implement a Python program that accepts three single-digit inputs (separated by spaces) and prints all six possible 3-digit combinations of these digits. The program must treat the inputs as strings to maintain any leading zeros through concatenation.

2. Pseudocode

1. **START**
2. **READ** a single line of input from the user, split it by spaces, and **STORE** the three string values into variables a, b, and c.
3. **PRINT** the concatenation of a, b, and c (representing the combination abc).
4. **PRINT** the concatenation of a, c, and b (representing the combination acb).
5. **PRINT** the concatenation of b, a, and c (representing the combination bac).
6. **PRINT** the concatenation of b, c, and a (representing the combination bca).
7. **PRINT** the concatenation of c, a, and b (representing the combination cab).
8. **PRINT** the concatenation of c, b, and a (representing the combination cba).
9. **END**

3. Flowchart



4. Python Program

```
a, b, c = input().split()
```

```
print(a + b + c)
```

```
print(a + c + b)
```

```
print(b + a + c)
```

```
print(b + c + a)
```

```
print(c + a + b)
```

```
print(c + b + a)
```

5. Experiment Screenshot

The screenshot displays the CODETANTRA online programming interface. On the left, the problem statement for '7.1.2: Possible Combinations from Three Digits' is shown. It asks for a Python program to accept three digits and print all possible 3-digit combinations. The input format is a single line with three space-separated non-negative integers. The output format requires printing all possible 3-digit combinations, each on a new line, maintaining leading zeros. Constraints specify that digits are between 0 and 9 and can be repeated.

The code editor on the right contains the following Python code:

```
1 a, b, c = input().split()
2
3 print(a + b + c)
4 print(a + c + b)
5 print(b + a + c)
6 print(b + c + a)
7 print(c + a + b)
8 print(c + b + a)
```

Below the code editor, the execution results are shown. The average time is 0.003 s and the maximum time is 0.004 s. The output shows that 3 out of 3 shown test cases passed and 4 out of 4 hidden test cases passed. The expected output for Test case 1 is listed as 123, 123, 132, 213, 231, and 312.